



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Assignment 3 – FALL 2025

Course Title:	Digital System Design	Course Code:	CPE 344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Program Name:	BS Computer Engineering		
Semester:	7 th	Batch:	FA22	Section:	A, B
Submission Date:	9 th December, 2025		Maximum Marks:	20	
Name:			Registration Number:		

Important Instructions / Guidelines:

- Do your own work, PLAGARISM will be graded as ZERO
- No late submission.

Question 1:

[CLO2-PLO3-C5]

[10 Marks]

Use the Implication Table method to optimize the Sequential Logic by reducing the number of states for the FSM shown in Figure 1 and design its circuit at gate level. Briefly summarize the optimization you obtained.

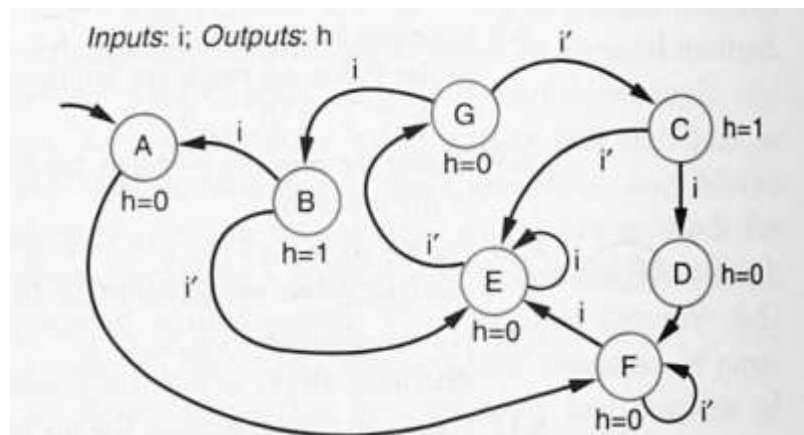


Figure 1

Question 2:

[CLO2-PLO3-C5]

[10 Marks]

Design a 16-bit multiplier using the sequential (add-and-shift) approach along with its controller. Compare your design with the array-style multiplier and clearly list the advantages and disadvantages of your proposed design.